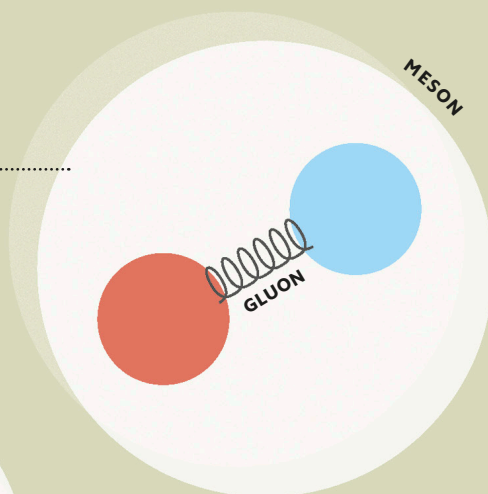
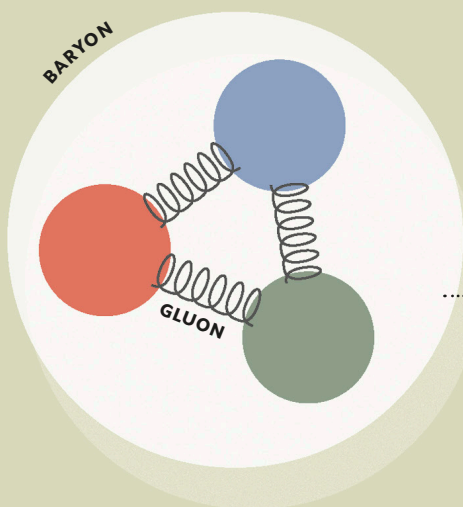


MESON (PAIR OF QUARKS)

A meson is made up of a quark-antiquark pair (see pp.128–129), bound by the strong force. The quarks have opposite color charge, resulting in an overall colorless particle.



BARYON



BARYON (ODD NUMBER OF QUARKS)

A baryon is made up of three quarks. For example, protons consist of two up quarks and one down quark bound by the strong force. All three primary colors (red, blue, and green) are represented, making it "colorless" overall.

THREE-COLOR QUARK

Quantum chromodynamics (QCD) is the quantum field theory for the strong interaction, which involves the exchange of gluons between quarks (see p.122). It has many parallels to QED (see opposite), with color instead of electric charge and gluons instead of photons.

However, the strong interaction has some unique characteristics, resulting in behavior such as color confinement (which means that color-charged particles cannot be found on their own) and an extremely limited range of approximately 10^{-15} m.